



Oulad Dahman Ilyass

📍 Salé, Morocco

🌐 [linkedin.com/in/ilyassod](https://www.linkedin.com/in/ilyassod)

🌐 Portfolio

✉ ilyassouladdahman@gmail.com

🐙 github.com/oudala

🔗 leetcode.com/u/SPOP00

About Me

Software and Cloud Engineer passionate about building scalable cloud-native applications and solving complex backend challenges. Experienced with distributed systems, observability, DevOps workflows, and modern software engineering practices.

Education

Engineering Degree in Computer Science

2023 – Present

National School of Applied Sciences, Al Hoceima

Specialized engineering program focused on software engineering, cloud computing, distributed systems, and computer science fundamentals.

Preparatory Classes for Engineering Cycle

2021 – 2023

National School of Applied Sciences, Al Hoceima

Completed intensive preparatory coursework in mathematics, algorithms, physics, and engineering fundamentals.

Professional Experience

Oracle – APM Session Diagnostics Engineering Intern (PFE Internship)

2026 – Present

Oracle Corporation

Contributed to Oracle APM session diagnostics and observability features, focusing on user session analytics, troubleshooting workflows, and platform diagnostics. Worked on backend improvements, ticket resolution, visualization systems, and enterprise monitoring support.

Technologies: Java, Python, Oracle Cloud, REST APIs, Git, CI/CD, Jira, Slack, Oracle Application Performance Monitoring (APM), Telemetry, Unit testing, Confluence, devops portal

Backend & Cloud Engineering Intern

Feb 2025 – Jun 2025

Univade LCNC Platform

Worked on a Low-Code/No-Code platform integrating backend services, automation pipelines, and cloud deployments. Developed microservices, CI/CD workflows, and infrastructure automation across AWS and Azure environments.

Technologies: Java, Python, Spring Boot, Docker, AWS, Azure, Terraform, GitHub Actions, Airflow

Technical Projects

Crescendo Baby Music Management Platform

2025 – Present

Developed a production-ready full-stack platform for managing children's music classes, bookings, subscriptions, attendance, payments, waitlists, scheduling, and admin operations. Implemented secure authentication, QR attendance tracking, WhatsApp notifications, analytics dashboards, and automation workflows.

booking.crescendobabymusic.com

Technologies: Next.js, React, TypeScript, Prisma, PostgreSQL, Supabase, Tailwind CSS, Twilio WhatsApp API, Cloudinary, Jest, Playwright, Vercel

AWS Cost Monitoring System – Intelligent Cloud Surveillance

2025

AWS cost monitoring system with anomaly detection and automated Slack alerts using a serverless architecture for intelligent cloud cost surveillance.

Technologies: AWS Lambda, S3, EventBridge, Python, Next.js, TypeScript, Tailwind CSS, Gemini AI

Création d'un Cloud Privé avec Apache CloudStack **2025**

Deployed a full private cloud using Apache CloudStack, including zones, pods, clusters, storage, and virtual networking. Automated VM provisioning using Terraform and Bash, with monitoring through Prometheus and Grafana.

Technologies: CloudStack, KVM, Linux, Networking, Prometheus, Grafana, Bash

High-Performance Persistent HashMap **2025**

Built a custom persistent HashMap reducing memory usage by 39% with optimized binary serialization, achieving 28x faster serialization performance. Used VisualVM and JMH for profiling and performance tuning.

Technologies: Java, Custom Serialization, Concurrency, VisualVM, JMH

MyORM – Custom Java ORM Library **2024**

Built a lightweight ORM using Java Reflection and Annotations for database mapping. Implemented a caching layer and JDBC-based query engine.

Technologies: Java, Reflection, JDBC, Caffeine Cache, JUnit

Custom ThreadPool Executor **2024**

Implemented a custom thread pool executor with a blocking queue and adjustable execution policies, designed for improved concurrency performance and controlled workload management.

Technologies: Java, Concurrency, Multithreading

Entrepreneurship & Social Impact

MamaFood – Traditional Food Delivery Platform **2024**

Co-founded a digital food delivery platform focused on connecting local traditional food providers with customers while optimizing logistics and operational workflows.

Technologies: Flask, Docker, Plotly, PostgreSQL

Morocco Water Race Competition **2024**

Designed an IoT-based water management prototype with monitoring dashboards and analytics systems for water resource optimization.

Technologies: Python, PostgreSQL, IoT Systems

Skills & Technologies

Programming Languages: Python, Java, C++, JavaScript, SQL, PL/SQL

Cloud & DevOps: AWS, Azure, OCI, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins

Frontend Technologies: Next.js, React, TypeScript, Tailwind CSS, Radix UI

Backend Technologies: Spring Boot, Flask, REST APIs, Microservices

Databases: MySQL, PostgreSQL, MongoDB, Redis

Testing & Monitoring: Pytest, Selenium, SonarQube, Prometheus, Grafana

Tools & Collaboration: Git, Slack, Jira, Bitbucket, Linux, VS Code, IntelliJ, Maven

Specialization: System Design, Distributed Systems, Performance Optimization

Certifications

Oracle Certified Professional: Java SE 17 Developer (OCP 17)

Professional Java certification covering advanced Java programming, concurrency, streams, collections, and enterprise development concepts.

Oracle Certified Professional: Java SE 11 Developer (OCP 11)

Validated expertise in Java application development, functional programming, modularity, and modern Java ecosystem practices.

Oracle Cloud Infrastructure 2024 Foundations Associate

Certification covering OCI architecture, networking, cloud services, security, and infrastructure deployment fundamentals.

AWS Cloud Practitioner Essentials – Ongoing

Cloud certification pathway focused on AWS services, cloud architecture, and deployment models.

Leadership & Activities

Enactus ENSA Al Hoceima – President

Led entrepreneurship and innovation initiatives, coordinated multidisciplinary teams, and managed social-impact technology projects and competitions.

Interests

AI, Cloud Computing, Competitive Programming, Open Source, System Architecture